4.1

If an ideal $\mathfrak{a}$ has a primary decomposition, then $\mathrm{Spec}(A/\mathfrak{a})$ has only finitely many irreducible components.
Proof: Every irreducible component of $\mathrm{Spec}(A/\mathfrak{a})$ is in the form $V(\pi(\mathfrak{p}))$ where $\pi(\mathfrak{p})$ is a minimal prime ideal in $A/\mathfrak{a}$, so $\mathfrak{p}$ is minimal in the prime ideals containing $\mathfrak{a}$. Since $\mathfrak{a}$ has a primary decomposition, there are only finitely many minimal ideals.

4.2

If $\mathfrak{a} = r(\mathfrak{a})$, then $\mathfrak{a}$ has no embedded prime ideals.
Proof: If $\mathfrak{a}$ has a primary decomposition, since $\mathfrak{a} = r(\mathfrak{a}) = \bigcap_{\mathfrak{p} \supset \mathfrak{a}, \mathfrak{p} \text{ minimal}} \mathfrak{p}$, where $\mathfrak{p}$ are $\mathfrak{p}-$primary, $\mathfrak{a}$ has no embedded prime ideals.

4.3

If A is absolutely flat, every primary ideal is maximal.
Proof: If $\mathfrak{q}$ is $\mathfrak{p}-$primary, by Exercise2.28 $k = A/\mathfrak{q}$ is absolutely flat, so every non-unit is a zero divisor, which is nilpotent since $\mathfrak{q}$ is primary. Hence k is an absolutely flat local ring, so k is a field and $\mathfrak{q}$ is maximal.

4.4

In the polynomial ring $\mathbb{Z}[t]$, the ideal $\mathfrak{m} = (2, t)$ is maximal and the ideal $\mathfrak{q} = (4, t)$ is $\mathfrak{m}$-primary, but is not a power of $\mathfrak{m}$.
Proof: $\mathbb{Z}[t]/\mathfrak{m} \cong \mathbb{Z}_2$ is a field, and $\mathbb{Z}[t]/\mathfrak{q} \cong \mathbb{Z}_4$ whose zero divisors $\bar{0}, \bar{2}$ are nilpotent, so $\mathfrak{m}$ is maximal and $\mathfrak{q}$ is primary. $\mathfrak{m} \subset r(\mathfrak{q})$ so $\mathfrak{q}$ is $\mathfrak{m}-$primary. However, $t \notin \mathfrak{m}^2$ so $\mathfrak{m}^2 \subsetneq \mathfrak{q} \subsetneq \mathfrak{m}$, and $\mathfrak{q}$ is not a power of $\mathfrak{m}$.

4.5

In the polynomial ring $K[x, y, z]$ where K is a field and x, y, z are independent indeterminates, let $\mathfrak{p}_1 = (x, y)$, $\mathfrak{p}_2 = (x, z)$, $\mathfrak{m} = (x, y, z)$; $\mathfrak{p}_1$ and $\mathfrak{p}_2$ are prime, and $\mathfrak{m}$ is maximal. Let $\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2$. Show that $\mathfrak{a} = \mathfrak{p}_1 \cap \mathfrak{p}_2 \cap \mathfrak{m}^2$ is a reduced primary decomposition of $\mathfrak{a}$. Which components are isolated and which are embedded?
Solution: Verify that $\mathfrak{p}_1, \mathfrak{p}_2 \subsetneq \mathfrak{m}$, while $\mathfrak{p}_1 \not\subset \mathfrak{p}_2 \not\subset \mathfrak{p}_1$, so $\mathfrak{p}_1, \mathfrak{p}_2$ are isolated and $\mathfrak{m}$ is embedded.

4.6

Let X be an infinite compact Hausdorff space, $C(X)$ the ring of real-valued continuous functions on X (Chapter 1, Exercise 26). Is the zero ideal decomposable in this ring?
Solution: No. For any $x \in X$, $r(a : x) = \{f : f(x) = 0\} = \mathfrak{m}_x$ is maximal hence prime, so they are all associated primes of 0, and there is no finite decomposition of 0.

4.7

Let A be a ring and let $A[x]$ denote the ring of polynomials in one indeterminate over A. For each ideal $\mathfrak{a}$ of A, let $\mathfrak{a}[x]$ denote the set of all polynomials in $A[x]$ with coefficients in $\mathfrak{a}$.
i) $\mathfrak{a}[x]$ is the extension of $\mathfrak{a}$ to $A[x]$.
ii) If $\mathfrak{p}$ is a prime ideal in A, then $\mathfrak{p}[x]$ is a prime ideal in $A[x]$.
iii) If $\mathfrak{q}$ is a $\mathfrak{p}$-primary ideal in A, then $\mathfrak{q}[x]$ is a $\mathfrak{p}[x]$-primary ideal in $A[x]$.
iv) If $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ is a minimal primary decomposition in A, then $\mathfrak{a}[x] = \bigcap_{i=1}^n \mathfrak{q}_i[x]$ is a minimal primary decomposition in $A[x]$.
v) If $\mathfrak{p}$ is a minimal prime ideal of $\mathfrak{a}$, then $\mathfrak{p}[x]$ is a minimal prime ideal of $\mathfrak{a}[x]$.
Proof: (i) is trivial. (ii) is Exercise2.7.
(iii) $A[x]/\mathfrak{q}[x] \cong (A/\mathfrak{q})[x]$. If $\sum \bar{a}_i x^i$ is a zero divisor, then $\bar{a}_i$ are zero divisors by Exercise1.2, hence they are nilpotent, and $\sum a_i x^i$ is nilpotent. Therefore $\mathfrak{q}[x]$ is primary.
Note that $\mathfrak{p}[x] \subset r(\mathfrak{q}[x])$, so $\mathfrak{q}[x]$ is $\mathfrak{p}[x]-$primary.
(iv) Clearly $\mathfrak{a}[x] = \bigcap_{i=1}^n \mathfrak{q}_i[x]$ and $q_i[x]$ are $p_i[x]-$primary. Since p_i are distinct, so are $\mathfrak{p}_i[x]$, and $\bigcap_{j\neq i} \mathfrak{q}_j \not\subset \mathfrak{q}_i$ implies $\bigcap_{j\neq i} \mathfrak{q}_j[x] \not\subset \mathfrak{q}_i[x]$. Hence it is a minimal decomposition.
(v) $\mathfrak{p}$ is minimal in $\mathfrak{a}$ implies $\mathfrak{p}^c$ is a minimal prime in $A/\mathfrak{a}$. Since prime ideals $\mathfrak{q}$ in $A/\mathfrak{a}$ correspond to prime ideals $\mathfrak{q}[x]$ in $(A/\mathfrak{a})[x]$, $\mathfrak{p}^c[x]$ is minimal in $(A/\mathfrak{a})[x] \cong A[x]/\mathfrak{a}[x]$, hence $\mathfrak{p}[x]$ is a minimal prime ideal of $\mathfrak{a}[x]$.

4.8

Let k be a field. Show that in the polynomial ring $k[x_1, \ldots, x_n]$ the ideals $\mathfrak{p}_i = (x_1, \ldots, x_i)$ $(1 \leqslant i \leqslant n)$ are prime and all their powers are primary.
Proof: $k[x_1, \cdots, x_n]/(x_1, \cdots, x_i) \cong k[x_{i+1}, \cdots, x_n]$ is an integral domain so $\mathfrak{p}_i$ is prime. We show that $\mathfrak{q}_i = \mathfrak{p}_i \cap k[x_1, \cdots, x_i]$ satisfy $\mathfrak{q}_i^m$ is primary, then $p_i^m = q_i^m[x_{i+1}, \cdots, x_n]$ is primary. So we can assume $i = n$.
Note that $k[x_1, \cdots, x_n]/(x_1, \cdots, x_n)^m = k[y_1, \cdots, y_n]$ where $y_{i_1} \cdots y_{i_m} = 0 \forall i_1, \cdots, i_m \leqslant n$. Every zero divisor has zero constant term, and y_i are all nilpotent, hence every zero divisor is nilpotent, and $(x_1, \cdots, x_n)^m$ is primary.

4.9

In a ring A, let $D(A)$ denote the set of prime ideals $\mathfrak{p}$ which satisfy the following condition: there exists $a \in A$ such that $\mathfrak{p}$ is minimal in the set of prime ideals containing $(0 : a)$. Show that $x \in A$ is a zero divisor $\iff$ $x \in \mathfrak{p}$ for some $\mathfrak{p} \in D(A)$.
Let S be a multiplicatively closed subset of A, and identify $\operatorname{Spec}(S^{-1}A)$ with its image in $\operatorname{Spec}(A)$ (Chapter 3, Exercise 21). Show that

$$D(S^{-1}A) = D(A) \cap \operatorname{Spec}(S^{-1}A).$$

If the zero ideal has a primary decomposition, show that $D(A)$ is the set of associated prime ideals of 0.
Proof: (i) If $x \in A$ is a zero divisor, suppose $xa = 0$, then $x \in (0 : a)$. Consider a minimal prime ideal $\mathfrak{p}$ containing $(0 : a)$ (such prime ideals always exist since minimal primes corresponds to maximal multiplicative sets, see Exercise1.8), then $x \in (0 : a) \subset \mathfrak{p}$ where $\mathfrak{p} \in D(A)$.
If $x \in \mathfrak{p}$ for some $\mathfrak{p} \in D(A)$ where $\mathfrak{p}$ is minimal containing $(0 : a)$, then $a \neq 0$ since $1 \in (0 : 0)$. $\bar{\mathfrak{p}}$ is a minimal prime in $A/(0 : a)$, so $S = A/(0 : \mathfrak{a}) - \bar{\mathfrak{p}}$ is a maximal multiplicative set without zero. Consider $S' = \{s\bar{x}^n : s \in S, n \geqslant 0\}$, then $S \subset S'$ and S' is multiplicative. Hence $0 \in S'$ so $s\bar{x}^n = 0$ for some $s \in S$

and $n > 0$. Take the minimal such n, let $\bar{y} = s\bar{x}^{n-1}$, then $\bar{y} \neq 0$ and $xy \in (0 : \mathfrak{a})$. Hence $(ay)x = 0$ for some $a \in \mathfrak{a}$. Since $\bar{y} \neq 0$, $y \notin (0 : \mathfrak{a})$ so $ay \neq 0$, hence x is a zero divisor.
(ii) Clearly $D(S^{-1}A) \subset S^{-1}X = \{\mathfrak{p} : \mathfrak{p} \cap S = \emptyset\}$.
For $x \in A$, we have $\mathrm{Ann}(x)^e = \left\{\frac{a}{s} : ax = 0\right\} = \mathrm{Ann}\left(\frac{x}{1}\right)$. Hence $S^{-1}\mathfrak{p} \in D(S^{-1}A)$ iff exists x such that $S^{-1}\mathfrak{p}$ is minimal containing $\mathrm{Ann}(x/1) = \mathrm{Ann}(x)^e$. Taking contractions we obtain $\mathfrak{p} \supset \mathrm{Ann}(x)$ and $\mathfrak{p}$ is minimal, therefore $D(S^{-1}A) \subset D(A)$. On the other hand, suppose $\mathfrak{p} \in D(A)$ and $\mathfrak{p} \cap S = \emptyset$, take $x \in A$ such that $\mathfrak{p}$ is minimal containing $\mathrm{Ann}(x)$, then $S^{-1}\mathfrak{p}$ is minimal containing $\mathrm{Ann}(x/1)$ so $S^{-1}\mathfrak{p} \in D(S^{-1}A)$. Therefore $D(S^{-1}A) = D(A) \cap \mathrm{Spec}(S^{-1}A)$.
(iii) Suppose $0 = \bigcap_{i=1}^n \mathfrak{q}_i$ has a primary decomposition, where $\mathfrak{q}_i$ is $\mathfrak{p}_i-$primary. Then $\mathfrak{p}_i$ are the prime ideals in $r(0 : x)$, so if $\mathfrak{p}_i = r(0 : x)$ it is the minimal prime ideal containing $(0 : x)$. If $\mathfrak{p}$ is minimal containing $(0 : x)$, then $\mathfrak{p} \supset r(0 : x) = \bigcap_{i=1}^n r(\mathfrak{q}_i : x) = \bigcap_{i=1}^m \mathfrak{p}_i$. Suppose $\mathfrak{p}_i \subset \mathfrak{p}$, then note that $\mathfrak{p}_i = r(\mathfrak{q}_i : x) \supset \mathfrak{p}$, hence $\mathfrak{p} = \mathfrak{p}_i$ is an associated prime.

4.10

For any prime ideal $\mathfrak{p}$ in a ring A, let $S_{\mathfrak{p}}(0)$ denote the kernel of the homomorphism $A \to A_{\mathfrak{p}}$. Prove that
i) $S_{\mathfrak{p}}(0) \subseteq \mathfrak{p}$.
ii) $r(S_{\mathfrak{p}}(0)) = \mathfrak{p} \Leftrightarrow \mathfrak{p}$ is a minimal prime ideal of A.
iii) If $\mathfrak{p} \supseteq \mathfrak{p}'$, then $S_{\mathfrak{p}}(0) \subseteq S_{\mathfrak{p}'}(0)$.
iv) $\bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0) = 0$, where $D(A)$ is defined in Exercise 9.
Proof: (i) If $a \in S_{\mathfrak{p}}(0)$ then $sa = 0$ for some $s \notin \mathfrak{p}$. Since $0 = sa \in \mathfrak{p}$, we have $a \in \mathfrak{p}$ and $S_{\mathfrak{p}}(0) \subset \mathfrak{p}$.
(ii) If $\mathfrak{q} \subsetneq \mathfrak{p}$ for some prime ideal $\mathfrak{q}$, then $sa = 0 \in \mathfrak{q}$ and $s \notin \mathfrak{p} \supset \mathfrak{q}$ implies $a \in \mathfrak{q}$. Hence $S_{\mathfrak{p}}(0) \subset \mathfrak{q}$ and $r(S_{\mathfrak{p}}(0)) \subset \mathfrak{q}$.
If $\mathfrak{p}$ is a minimal prime then $S = A - \mathfrak{p}$ is a maximal multiplicative set without zero, so for any $x \in \mathfrak{p}$, $sx^n = 0$ for some $s \in S, n > 0$. Hence $r(S_{\mathfrak{p}}(0)) = \mathfrak{p}$.
(iii) Note that $S_{\mathfrak{p}}(0) = \bigcup_{s \notin \mathfrak{p}} \mathrm{Ann}(s) \subset \bigcup_{s \notin \mathfrak{p}'} \mathrm{Ann}(s) = S_{\mathfrak{p}'}(0)$.
(iv) For any $a \neq 0$, $(0 : a) \neq (1)$ so take $\mathfrak{p} \in D(A)$ minimal in containing $(0 : a)$, then $(0 : a) \subset \mathfrak{p}$ implies $a \notin S_{\mathfrak{p}}(0)$. Hence $\bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0) = 0$.

4.11

If $\mathfrak{p}$ is a minimal prime ideal of a ring A, show that $S_{\mathfrak{p}}(0)$ (Exercise 10) is the smallest $\mathfrak{p}$-primary ideal.
Let $\mathfrak{a}$ be the intersection of the ideals $S_{\mathfrak{p}}(0)$ as $\mathfrak{p}$ runs through the minimal prime ideals of A. Show that $\mathfrak{a}$ is contained in the nilradical of A.
Suppose that the zero ideal is decomposable. Prove that $\mathfrak{a} = 0$ if and only if every prime ideal of 0 is isolated.
(i) By Ex4.10(ii) $r(S_{\mathfrak{p}}(0)) = \mathfrak{p}$. If $xy \in S_{\mathfrak{p}}(0)$ and $x \notin \mathfrak{p}$, then $sxy = 0$ for some $s \in S$, so $sx \in S$ and $(sx)y = 0$. Hence $y \in S_{\mathfrak{p}}(0)$ and $S_{\mathfrak{p}}(0)$ is $\mathfrak{p}-$primary. If $\mathfrak{q}$ is a $\mathfrak{p}-$primary ideal, then $sa = 0 \in \mathfrak{q}$ for $s \notin \mathfrak{p} = r(\mathfrak{q})$ implies $a \in \mathfrak{q}$ so $S_{\mathfrak{p}}(0) \subset \mathfrak{q}$.
(ii) Use Ex4.10(iii) and $S_{\mathfrak{p}}(0) \subset \mathfrak{p}$,

$$\mathfrak{a} = \bigcap_{\mathfrak{p} \text{ minimal}} S_{\mathfrak{p}}(0) = \bigcap_{\mathfrak{p} \in \mathrm{Spec} A} S_{\mathfrak{p}}(0) \subset \bigcap_{\mathfrak{p} \in \mathrm{Spec} A} \mathfrak{p} = \mathfrak{N}.$$

(iii) Suppose $0 = \bigcap_{i=1}^n \mathfrak{q}_i$ where $\mathfrak{q}_i$ are $\mathfrak{p}_i-$primary.
If $\mathfrak{p}_i$ are isolated, then $\mathfrak{a} = \bigcap_{\mathfrak{p}\in \mathrm{Spec} A} S_\mathfrak{p}(0) \subset \bigcap_{i=1}^n S_{\mathfrak{p}_i}(0) \subset \bigcap_{i=1}^n \mathfrak{q}_i = 0$ so $\mathfrak{a} = 0$.
If $\mathfrak{a} = 0$ suppose $q_1, \cdots, q_m$ does not meet $S_\mathfrak{p} = A - \mathfrak{p}$ then by Ex4.12(i) $S_\mathfrak{p}(0) = \bigcap_{i=1}^m \mathfrak{q}_i$ where $S_\mathfrak{p}(0)$ and $\mathfrak{q}_i$ are primary. Hence $\mathfrak{p} = r(S_\mathfrak{p}(0)) = \bigcap_{i=1}^m \mathfrak{p}_i$ so every minimal prime is one of $\mathfrak{p}_i$. Therefore $0 = \bigcap_{\mathfrak{p}\text{ minimal}} S_\mathfrak{p}(0)$ is a minimal primary decomposition, and all associated primes are isolated.

4.12

Let A be a ring, S a multiplicatively closed subset of A. For any ideal $\mathfrak{a}$, let $S(\mathfrak{a})$ denote the contraction of $S^{-1}\mathfrak{a}$ in A. The ideal $S(\mathfrak{a})$ is called the **saturation** of $\mathfrak{a}$ with respect to S. Prove that
i) $S(\mathfrak{a}) \cap S(\mathfrak{b}) = S(\mathfrak{a} \cap \mathfrak{b})$
ii) $S(r(\mathfrak{a})) = r(S(\mathfrak{a}))$
iii) $S(\mathfrak{a}) = (1) \Leftrightarrow \mathfrak{a}$ meets S
iv) $S_1(S_2(\mathfrak{a})) = (S_1S_2)(\mathfrak{a})$.
If $\mathfrak{a}$ has a primary decomposition, prove that the set of ideals $S(\mathfrak{a})$ (where S runs through all multiplicatively closed subsets of A) is finite.
Proof: (i) Clearly $S(\mathfrak{a} \cap \mathfrak{b}) \subset S(\mathfrak{a}), S(\mathfrak{b})$. If $c \in S(\mathfrak{a}) \cap S(\mathfrak{b})$, suppose $\frac{c}{1} = \frac{a}{s} = \frac{b}{t}$ for $a \in \mathfrak{a}, b \in \mathfrak{b}$, take $s' \in S$ such that $s'ta = s'sb \in \mathfrak{a} \cap \mathfrak{b}$, then $\frac{c}{1} = \frac{s'ta}{s'st} \in S^{-1}(\mathfrak{a} \cap \mathfrak{b})$ so $c \in S(\mathfrak{a} \cap \mathfrak{b})$.
(ii) Clearly $S(r(\mathfrak{a})) = r(a)^{ec} \subset r(a^e)^c = r(a^{ec}) = r(S(\mathfrak{a}))$.
If $\frac{r^n}{1} = \frac{a}{s} \in S^{-1}\mathfrak{a}$ where $a \in \mathfrak{a}$, take $s'sr^n = s'a \in \mathfrak{a}$ for $s' \in S$, then $\frac{r}{1} = \frac{s'sr}{s's} \in S^{-1}r(\mathfrak{a})$ so $r(S(\mathfrak{a})) \subset S(r(\mathfrak{a}))$.
(iii) If $s \in \mathfrak{a} \cap S$ then $1/1 = s/s \in S^{-1}\mathfrak{a}$ so $S(\mathfrak{a}) = (1)$.
If $S(\mathfrak{a}) = (1)$ then $\frac{1}{1} = \frac{a}{s}$ for some $a \in \mathfrak{a}$, so $s'a = s's \in \mathfrak{a} \cap S$ for some $s' \in S$>
(iv) If $x \in (S_1S_2)(\mathfrak{a})$ then $\exists s_1, s_2 \in S_1, S_2$ such that $s_1s_2x \in \mathfrak{a}$, so $s_1x \in S_2(\mathfrak{a})$ and $x \in S_1(S_2(\mathfrak{a}))$. The converse is analogous.
(v) Use Prop4.9, there are at most 2^n possible choices.

4.13

Let A be a ring and $\mathfrak{p}$ a prime ideal of A. The **nth symbolic power** of $\mathfrak{p}$ is defined to be the ideal (in the notation of Exercise 12)

$$\mathfrak{p}^{(n)} = S_\mathfrak{p}(\mathfrak{p}^n)$$

where $S_\mathfrak{p} = A - \mathfrak{p}$. Show that
i) $\mathfrak{p}^{(n)}$ is a $\mathfrak{p}$-primary ideal;
ii) if $\mathfrak{p}^n$ has a primary decomposition, then $\mathfrak{p}^{(n)}$ is its $\mathfrak{p}$-primary component;
iii) if $\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}$ has a primary decomposition, then $\mathfrak{p}^{(m+n)}$ is its $\mathfrak{p}$-primary component;
iv) $\mathfrak{p}^{(n)} = \mathfrak{p}^n \Leftrightarrow \mathfrak{p}^n$ is $\mathfrak{p}$-primary.
Proof: (i) Since $\mathfrak{p}-$primary ideals $\mathfrak{q}$ containing $\mathfrak{a}$ and $\mathfrak{p}/\mathfrak{a}-$primary ideals $\mathfrak{q}/\mathfrak{a}$ correspond bijectively, by Ex4.11(i) $\mathfrak{p}^{(n)}$ is the minimal $\mathfrak{p}-$primary ideal containing $\mathfrak{p}^n$.
(ii) If $\mathfrak{p}^n = \bigcap_{i=1}^m \mathfrak{q}_i$ and $S_\mathfrak{p}$ does not meet $\mathfrak{q}_1, \cdots, \mathfrak{q}_m$, then $\mathfrak{p}^{(n)} = S_\mathfrak{p}(\mathfrak{p}^n) = \bigcap_{i=1}^m \mathfrak{q}_i$ so $\mathfrak{p}^n = \mathfrak{p}^{(n)} \cap \left(\bigcap_{i=m+1}^n \mathfrak{q}_i\right)$ is a primary decomposition and $\mathfrak{p}^{(n)}$ is its $\mathfrak{p}-$primary component.
(iii) Likewise, $\mathfrak{p}^{(n+m)} = S_\mathfrak{p}(\mathfrak{p}^{n+m}) \subset S_\mathfrak{p}(\mathfrak{p}^{(n)}\mathfrak{p}^{(m)})$, and verify that $\mathfrak{p}^{(n)}\mathfrak{p}^{(m)} \subset \mathfrak{p}^{(n+m)}$ so

$S_{\mathfrak{p}}(\mathfrak{p}^{(n)}\mathfrak{p}^{(m)}) \subset S_{\mathfrak{p}}(\mathfrak{p}^{(n+m)}) = \mathfrak{p}^{(n+m)}$. Hence $\mathfrak{p}^{(n+m)} = S_{\mathfrak{p}}(\mathfrak{p}^{(n)}\mathfrak{p}^{(m)})$. Analogous to (ii) $\mathfrak{p}^{(n+m)}$ is its $\mathfrak{p}-$primary component,
(iv) Trivial by minimality of $\mathfrak{p}^{(n)}$.

4.14

Let $\mathfrak{a}$ be a decomposable ideal in a ring A and let $\mathfrak{p}$ be a maximal element of the set of ideals $(\mathfrak{a} : x)$, where $x \in A$ and $x \notin \mathfrak{a}$. Show that $\mathfrak{p}$ is a prime ideal belonging to $\mathfrak{a}$.
Proof: Suppose $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ where $\mathfrak{q}_i$ are $\mathfrak{p}_i-$primary, and $\mathfrak{p} = (\mathfrak{a} : x)$ is maximal. Suppose $yz \in \mathfrak{p}$ while $y \notin \mathfrak{p}$, then by $y \notin \mathfrak{p}$ we have $xy \notin \mathfrak{a}$, so $(\mathfrak{a} : x) \subset ((\mathfrak{a} : x) : y) = (\mathfrak{a} : xy)$ implies $z \in (\mathfrak{a} : xy) = (\mathfrak{a} : x)$. Hence $\mathfrak{p}$ is prime.
Note that $\mathfrak{p} = (\mathfrak{a} : x) = \bigcap_{i=1}^n (\mathfrak{q}_i : x)$, so $\mathfrak{p} = r(\mathfrak{a} : x) = \bigcap_{i=1}^n r(\mathfrak{q}_i : x) = \bigcap_{i=1}^m \mathfrak{p}_i$, assuming $x \notin \mathfrak{q}_1, \cdots, \mathfrak{q}_m$. Hence $\mathfrak{p} = \mathfrak{p}_i$ for some $1 \leqslant i \leqslant n$, and $\mathfrak{p}$ is a prime belonging to $\mathfrak{a}$.

4.15

Let $\mathfrak{a}$ be a decomposable ideal in a ring A, let Σ be an isolated set of prime ideals belonging to $\mathfrak{a}$, and let $\mathfrak{q}_\Sigma$ be the intersection of the corresponding primary components. Let f be an element of A such that, for each prime ideal $\mathfrak{p}$ belonging to $\mathfrak{a}$, we have $f \in \mathfrak{p} \iff \mathfrak{p} \notin \Sigma$, and let S_f be the set of all powers of f. Show that $\mathfrak{q}_\Sigma = S_f(\mathfrak{a}) = (\mathfrak{a} : f^n)$ for all large n.
Proof: $\Sigma = \{\mathfrak{p} \text{ associated} : f \notin \mathfrak{p}\}$. Suppose $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ where $\mathfrak{q}_i$ are $\mathfrak{p}_i-$primary and $f \notin \mathfrak{p}_1, \cdots, \mathfrak{p}_n$, then $S_f(\mathfrak{a}) = \bigcap_{i=1}^m \mathfrak{q}_i = \mathfrak{q}_\Sigma$ and $(\mathfrak{a} : f^n) = \bigcap_{i=1}^n (\mathfrak{q}_i : f^n) = \bigcap_{i=1}^m \mathfrak{q}_i = \mathfrak{q}_\Sigma$.

4.16

If A is a ring in which every ideal has a primary decomposition, show that every ring of fractions $S^{-1}A$ has the same property.
Proof: For any ideal $\mathfrak{b} \subset S^{-1}A$, let $\mathfrak{a} = \mathfrak{b}^c$ and take its primary decomposition $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ where $\mathfrak{q}_i$ are $\mathfrak{p}_i-$primary. Suppose S does not meet $\mathfrak{p}_1, \cdots, \mathfrak{p}_m$, then $S^{-1}\mathfrak{a} = \bigcap_{i=1}^m S^{-1}\mathfrak{q}_i$ where $S^{-1}\mathfrak{q}_i$ are $S^{-1}\mathfrak{p}_i-$primary. Since every ideal in $S^{-1}A$ is extended, $\mathfrak{b} = S^{-1}\mathfrak{a}$ has a primary decomposition.

4.17

Let A be a ring with the following property.
(L1) For every ideal $\mathfrak{a} \neq (1)$ in A and every prime ideal $\mathfrak{p}$, there exists $x \notin \mathfrak{p}$ such that $S_{\mathfrak{p}}(\mathfrak{a}) = (\mathfrak{a} : x)$, where $S_{\mathfrak{p}} = A - \mathfrak{p}$.
Then every ideal in A is an intersection of (possibly infinitely many) primary ideals.
Proof: Verify that L1 holds when passing to $A/\mathfrak{a}$, hence it suffices to prove $0 = \bigcap_{\mathfrak{q}} \mathfrak{q}$, $\mathfrak{q}$ taking all primary ideals. Recall that 0 is the intersection of all irreducible ideals (by applying Zorn's lemma to all ideals without $a \neq 0$). We show that if L1 holds, then every irreducible ideal is primary. Likewise passing to $A/\mathfrak{a}$ we need to prove if 0 is irreducible then 0 is primary.
Clearly $r(0) = \mathfrak{N}$ is the nilradical. If 0 is irreducible, then $\mathfrak{a} \cap \mathfrak{b} \neq 0$ for all $\mathfrak{a}, \mathfrak{b} \neq 0$. For any prime ideal $\mathfrak{p}$, take $x \notin \mathfrak{p}$ such that $S_{\mathfrak{p}}(0) = (0 : x)$, we show that $(x) \cap S_{\mathfrak{p}}(0) = 0$: If $y = ax \in (x) \cap S_{\mathfrak{p}}(0)$, then $x^2 \in S_{\mathfrak{p}}$ and $ax^2 = 0$ implies $a \in S_{\mathfrak{p}}(0) = (0 : x)$, so $y = ax = 0$.
Since 0 is irreducible and $x \neq 0 \implies (x) \neq 0$, we infer $S_{\mathfrak{p}}(0) = 0$ for all $\mathfrak{p}$ prime. Hence if $xy = 0$ and

$x \notin r(0) = \mathfrak{N}$, take $\mathfrak{p}$ prime such that $x \notin \mathfrak{p}$, then $y \in S_{\mathfrak{p}}(0) = 0$ so $y = 0$. Therefore 0 is primary.

4.18

Consider the following condition on a ring A:
(L2) Given an ideal $\mathfrak{a}$ and a descending chain $S_1 \supseteq S_2 \supseteq \cdots \supseteq S_n \supseteq \ldots$ of multiplicatively closed subsets of A, there exists an integer n such that $S_n(\mathfrak{a}) = S_{n+1}(\mathfrak{a}) = \ldots$. Prove that the following are equivalent:
i) Every ideal in A has a primary decomposition;
ii) A satisfies (L1) and (L2).
Proof: (i)=>(L1): If $\mathfrak{a} \neq (1)$ and $\mathfrak{p}$ is prime, consider the primary decomposition $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ where $\mathfrak{q}_i$ is $\mathfrak{p}_i-$primary. Suppose $S_{\mathfrak{p}}$ does not meet $\mathfrak{p}_1, \cdots, \mathfrak{p}_m$, then $S_{\mathfrak{p}}(\mathfrak{a}) = \bigcap_{i=1}^m \mathfrak{q}_i$, where $\mathfrak{p}_i \subset \mathfrak{p} \forall 1 \leqslant i \leqslant m$ and $\mathfrak{p}_i \not\subset \mathfrak{p}$. Note that $\bigcap_{i=m+1}^n \mathfrak{q}_i \not\subset \mathfrak{p}$, otherwise $\bigcap_{i=m+1}^n \mathfrak{p}_i \subset r(\mathfrak{p}) = \mathfrak{p}$ so $\mathfrak{p}_i \subset \mathfrak{p}$ for some $i > m$, a contradiction. Take $x \in (\bigcap_{i=m+1}^n \mathfrak{q}_i) \backslash \mathfrak{p}$, then $x \notin \mathfrak{p}_1, \cdots, \mathfrak{p}_m$ so $(\mathfrak{a} : x) = \bigcap_{i=1}^n (\mathfrak{q}_i : x) = \bigcap_{i=1}^m \mathfrak{q}_i = S_{\mathfrak{p}}(\mathfrak{a})$.
(i)=>(L2) Suppose $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ with $p_i = r(\mathfrak{q}_i)$. Then $S_n(\mathfrak{a}) = \bigcap_{i \in J_n} \mathfrak{q}_i$ where $J_n = \{i : \mathfrak{p}_i \cap S_n = \emptyset\}$. Since $J_1 \subset J_2 \subset \cdots$ and J_1 is finite, it will be stationary, and so it the sequence S_n.
(L1+L2)=>(i): Given an ideal $\mathfrak{a} \neq (1)$, by Ex4.17 $\mathfrak{a} = \bigcap_{\alpha \in J} \mathfrak{q}_\alpha$ where $\mathfrak{q}_\alpha$ are primary. Since for $p-$primary ideals $\mathfrak{q}_1, \mathfrak{q}_2$, $\mathfrak{q}_1 \cap \mathfrak{q}_2$ is still $\mathfrak{p}-$primary, we can suppose $r(\mathfrak{q}_\alpha)$ are distinct. Assume otherwise that $\mathfrak{a}$ has no finite primary decomposition. Then we can take a sequence $\mathfrak{q}_1, \cdots, \mathfrak{q}_n, \cdots$ with $\mathfrak{p}_i = r(\mathfrak{q}_i)$, such that $\mathfrak{q}_{n+1} \not\subset \bigcap_{i=1}^n \mathfrak{q}_i$. Let $S_n = S_{\mathfrak{p}_1} \cap \cdots \cap S_{\mathfrak{p}_n}$, then $S_1 \supset S_2 \supset \cdots$ is a descending chain of multiplicative sets. Then $S_n(a) = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$ is a non-stationary chain, a contradiction to (L2).

4.19

Let A be a ring and $\mathfrak{p}$ a prime ideal of A. Recall that every $\mathfrak{p}$-primary ideal contains $S_{\mathfrak{p}}(0)$, the kernel of the canonical homomorphism $A \to A_{\mathfrak{p}}$.
Suppose that A satisfies the following condition: for every prime ideal $\mathfrak{p}$, the intersection of all $\mathfrak{p}$-primary ideals of A is equal to $S_{\mathfrak{p}}(0)$. (Noetherian rings satisfy this condition: see Chapter 10.) Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ be distinct prime ideals, none of which is a minimal prime ideal of A. Then there exists an ideal $\mathfrak{a}$ in A whose associated prime ideals are $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$.
Proof: Proof by induction on n. The case $n = 1$ is trivial by taking $\mathfrak{a} = \mathfrak{p}_1$. Suppose $n > 1$ and let $\mathfrak{p}_n$ be maximal in the set $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$. By the inductive hypothesis there exists an ideal $\mathfrak{b}$ and a minimal primary decomposition $\mathfrak{b} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_{n-1}$, where each $\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary. If $\mathfrak{b} \subseteq S_{\mathfrak{p}_n}(0)$, let $\mathfrak{p}$ be a minimal prime ideal of A contained in $\mathfrak{p}_n$. Then $S_{\mathfrak{p}_n}(0) \subseteq S_{\mathfrak{p}}(0)$, hence $\mathfrak{b} \subseteq S_{\mathfrak{p}}(0)$. Taking radicals and using Exercise 10, we have $\mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_{n-1} \subseteq \mathfrak{p}$, hence some $\mathfrak{p}_i \subseteq \mathfrak{p}$, hence $\mathfrak{p}_i = \mathfrak{p}$ since $\mathfrak{p}$ is minimal. This is a contradiction since no $\mathfrak{p}_i$ is minimal.
Hence $\mathfrak{b} \nsubseteq S_{\mathfrak{p}_n}(0)$ and therefore there exists a $\mathfrak{p}_n$-primary ideal $\mathfrak{q}_n$ such that $\mathfrak{b} \nsubseteq \mathfrak{q}_n$. Consider $\mathfrak{a} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$ which is a primary decomposition of $\mathfrak{a}$ since $r(\mathfrak{q}_i) = \mathfrak{p}_i$ are distinct and $\mathfrak{b} = \bigcap_{i=1}^{n-1} \mathfrak{q}_i \not\subset \mathfrak{q}_n$. We show that $\mathfrak{b}_i \cap \mathfrak{q}_n \not\subset \mathfrak{q}_i$ where $\mathfrak{b}_i = \bigcap_{j \neq i,n} \mathfrak{q}_j$ by localizing at $\mathfrak{p}_i$: Since $\mathfrak{p}_n$ is maximal, $\mathfrak{p}_n \not\subset \mathfrak{p}_i$ so $S_{\mathfrak{p}_i}(\mathfrak{q}_n) = (1)$, hence by $\mathfrak{b}_i \not\subset \mathfrak{q}_i$ we have $\mathfrak{b}_i \cap \mathfrak{q}_n \not\subset \mathfrak{q}_n$. Therefore the associated primes of $\mathfrak{a}$ are $\mathfrak{p}_1, \cdots, \mathfrak{p}_n$.

Primary decomposition of modules
Practically the whole of this chapter can be transposed to the context of modules over a ring A. The following exercises indicate how this is done.

4.20

Let M be a fixed A-module, N a submodule of M. The **radical of** N **in** M is defined to be
$r_M(N) = \{x \in A : x^q M \subseteq N \text{ for some } q > 0\}$.
Show that $r_M(N) = r(N : M) = r(\mathrm{Ann}(M/N))$. In particular, $r_M(N)$ is an ideal.
State and prove the formulas for r_M analogous to (1.13).
Proof: (i) By definition $r_M(N) = r(\{y \in A : yM \subset N\}) = r(N : M) = r(\mathrm{Ann}(M/N))$, since
$yM \subset N \iff y(M/N) = 0$.
(ii) $r_M(N \cap N') = \{x : x^q M \subset N \cap N'\} = r_M(N) \cap r_M(N')$.
$r_M(N) = (1) \iff N = M$.
$r_M(N + N') \supset r(r_M(N) + r_M(N'))$, but the converse is false: Let $M = A \oplus A$ with action
$a(b, c) = (ab, ac)$, and $N = (0) \oplus A, N' = A \oplus (0)$, then $r_M(N + N') = (1)$ but
$r_M(N) = r_M(N') = (0)$.

4.21

An element $x \in A$ defines an endomorphism ϕ_x of M, namely $m \mapsto xm$. The element x is said to be a **zero-divisor** (resp. **nilpotent**) in M if ϕ_x is not injective (resp. is nilpotent). A submodule Q of M is **primary** in M if $Q \neq M$ and every zero-divisor in M/Q is nilpotent.
Show that if Q is primary in M, then $(Q : M)$ is a primary ideal and hence $r_M(Q)$ is a prime ideal $\mathfrak{p}$. We say that Q is $\mathfrak{p}$-**primary** (in M).
Prove the analogues of (4.3) and (4.4).
Proof: (i) Let $N = M/Q$ be primary, we show that $(Q : M) = \mathrm{Ann}(N)$ is primary. For any $xy \in \mathrm{Ann}(N)$ with $y \notin \mathrm{Ann}(N)$, take $n \in N$ such that $yn \neq 0$, then $xyn = 0$ implies $x(yn) = 0$ so x is a zero divisor in N. Since N is primary, x is a zero divisor in N hence $x^n \in \mathrm{Ann}(N)$ for some $n \geqslant 1$.
(ii)
Analogue of (4.3): If Q_i are $\mathfrak{p}-$primary, then $Q = \bigcap_{i=1}^n Q_i$ is $\mathfrak{p}-$primary.
Proof: $r_M(Q) = \bigcap_{i=1}^n r_M(Q_i) = \mathfrak{p}$. Let $xy \in Q$ where $y \notin Q$, then $xy \in Q_i$ for all $1 \leqslant i \leqslant n$. For some $1 \leqslant i \leqslant n$, $y \notin Q_i$ and $xy \in Q_i$, hence $x \in \mathfrak{p}$, and Q is $\mathfrak{p}-$primary.
Analogue of (4.4): Let Q be a $\mathfrak{p}-$primary ideal, then (a) if $x \in Q$ then $(Q : x) = (1)$; (b) if $x \notin Q$ then $(Q : x)$ is $\mathfrak{p}-$primary; (c) if $x \in A$ and $x \notin \mathfrak{p}$, then $(Q : x) = \{m \in M : xm \in Q\} = Q$.
Proof: (a) By definition $1 \cdot x \in Q$ so $(Q : x) = (1)$.
(c) If $xm \in Q$ with $m \notin Q$, then $x\bar{m} = 0$ with $\bar{m} \neq 0$ so x is a zero divisor. Hence $x \in \mathfrak{p}$ a contradiction. Therefore $(Q : x) = Q$.
(b) If $ab \in (Q : x)$ and $b \notin (Q : x)$, then $abx \in Q$ with $bx \notin Q$. Hence a is a zero-divisor in M/Q, so $a^n \bar{x} = 0$ for some $n \geqslant 1$, which implies $a \in r(Q : x)$ and $(Q : x)$ is primary.
Since $(Q : M) \subset (Q : x) \subset \mathfrak{p}$, we have $\mathfrak{p} = r(Q : M) \subset r(Q : x) \subset \mathfrak{p}$. Therefore $(Q : x)$ is $\mathfrak{p}-$primary.

4.22

A **primary decomposition** of N in M is a representation of N as an intersection
$N = Q_1 \cap \cdots \cap Q_n$
of primary submodules of M; it is a **minimal primary decomposition** if the ideals $\mathfrak{p}_i = r_M(Q_i)$ are all distinct and if none of the components Q_i can be omitted from the intersection, that is if $Q_i \nsubseteq \bigcap_{j \neq i} Q_j$ (

$1 \leqslant i \leqslant n$).
Prove the analogue of (4.5), that the prime ideals $\mathfrak{p}_i$ depend only on N (and M). They are called the **prime ideals belonging to N in M**. Show that they are also the prime ideals belonging to 0 in M/N.
Proof: (i) We show that the set of primes $\mathfrak{p}_i$ are exactly the prime ideals occurring in $\{r(N:m) : m \in M\}$. Note that if we assume $m \notin Q_1, \cdots, Q_k$, then $r(N:m) = \bigcap_{i=1}^n r(Q_i : m) = \bigcap_{i=1}^k \mathfrak{p}_i$. Hence if $\mathfrak{p} = r(N:m)$ is prime, then $\mathfrak{p} = \mathfrak{p}_i$ for some $1 \leqslant i \leqslant n$. Conversely, for each $1 \leqslant i \leqslant n$ there exists $m \in \bigcap_{j \neq i} Q_j \backslash Q_i$, then $r(N:m) = \mathfrak{p}_i$.
(ii) Trivial since $r(N:m) = r(0:\bar{m})$.

4.23

State and prove the analogues of (4.6)-(4.11) inclusive. (There is no loss of generality in taking $N = 0$.)
(4.6'): Let N be a decomposable submodule, then any prime ideal $\mathfrak{p} \supset r_M(N)$ contains a minimal prime ideal belonging to N, and thus the minimal prime ideals of N are precisely the minimal elements in the set of all prime ideals containing $r_M(N)$.
Proof: If $\mathfrak{p} \supset r_M(N) = \bigcap_{i=1}^n \mathfrak{p}_i$, then $\mathfrak{p} \supset \mathfrak{p}_i$ for some i, so $\mathfrak{p}$ contains some minimal prime ideal of N.
(4.7'): Let N be a decomposable submodule, and $N = \bigcap_{i=1}^n Q_i$ be a minimal primary decomposition, with $r_M(Q_i) = \mathfrak{p}_i$. Then

$$\bigcup_{i=1}^n \mathfrak{p}_i = \{x \in A : (N:x) \neq N\}.$$

In particular, if 0 is decomposable, the set $D \subset A$ of zero-divisors of A is the union of prime ideals belonging to 0.
Proof: It suffices to prove the case $N = 0$. Note that $D = r(D)$ since $x^n m = 0$ for some $m \neq 0$ implies $x \cdot (x^{n-1}m) = 0$ so taking the least n we have $x \in D$. By definition $D = \bigcup_{m \neq 0}(0:m)$, so $D = \bigcup_{m \neq 0} r(0:m)$. Since each $r(0:m)$ is the intersection of $\mathfrak{p}_i$, and $\exists m, r(0:m) = \mathfrak{p}_i$ for all i. Hence $D = \bigcup_{i=1}^n \mathfrak{p}_i$.
(4.8'): Let S be a multiplicative subset of A, and Q be a $\mathfrak{p}-$primary submodule. (a) If $S \cap \mathfrak{p} \neq \emptyset$, then $S^{-1}Q = S^{-1}M$; (b) If $S \cap \mathfrak{p} = \emptyset$, then $S^{-1}Q$ is $S^{-1}\mathfrak{p}-$primary and its contraction in M is Q. Hence the primary submodules in $S^{-1}M$ correspond to primary submodules in M.
Proof: (a) If $s \in S \cap \mathfrak{p} \neq \emptyset$, then for some $n \geqslant 1$, $s^n \in S \cap (Q:M)$. Hence for any $\frac{m}{s'} \in S^{-1}M$, $\frac{m}{s'} = \frac{s^n m}{s^n s'} \in S^{-1}Q$, so $S^{-1}Q = S^{-1}M$.
(b) If $S \cap \mathfrak{p} = \emptyset$, and $\frac{a}{s} \cdot \frac{b}{t} \in S^{-1}Q$ with $\frac{b}{t} \notin S^{-1}Q$, then $s'(rab - stq) = 0$ for some $s', r \in S$ and $q \in Q$. Hence $s'rab = s'stq \in Q$, with $s'rb \notin Q$ since $\frac{b}{t} \notin S^{-1}Q$. Therefore $a \in \mathfrak{p}$, so $\frac{a}{s} \in S^{-1}\mathfrak{p}$ and $S^{-1}Q$ is $S^{-1}\mathfrak{p}-$primary. It is easy to verify $(S^{-1}Q)^c = Q$.
(Skip the rest).